



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

February 15, 2024

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 21290
(THIRD REVISION)

EXPIRATION DATE: 2027-05-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Orion Engineered Carbons LLC
Spring, TX
2. PURPOSE AND LIMITATION:
 - a. This special permit authorizes the filling and transportation in commerce, for export only, of dinitrogen tetroxide in non-DOT specification pressure drums. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in the development of this special permit only considered the hazards and risks associated with the transportation in commerce.
 - c. No party status will be granted to this special permit.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR §§ 171.23(a)(1) and 171.23(b)(10) and § 173.314 in that non-DOT specification pressure vessels are authorized as provided herein.
5. BASIS: This special permit is based on the application of Orion Engineered Carbons LLC dated June 28, 2023, in accordance with § 107.105 and the public proceeding thereon and additional information dated January 31, 2024.

Tracking Number: 2024015731

6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Dinitrogen tetroxide	2.3	UN1067	N/A Hazard Zone A

7. SAFETY CONTROL MEASURES:a. PACKAGING:

(1) Prescribed packaging is non-DOT specification welded steel drum, which is designed, manufactured, and tested in accordance with UN ISO 21172-1: 2015 standard by CHEMET S. A. (Poland) and is UN stamp marked. The pressure drum is also Pi marked in accordance with the European Directive 2010/35/EU on transportable pressure equipment directive (TPED). The pressure drum must be in conformance with CHEMET S. A. drawings No. 3-1855-01/K Rev.0, 1-1855-01 Rev.1 and strength calculations No. 4-1855-02.2 on file with the Office of Hazardous Materials Safety (OHMS).

(2) Each pressure drum must conform to the following design data:

Steel for shell and dished ends	P355NL1 (Carbon-Mn steel) ¹
Water capacity	880L (31.0 cu ft)
Outside diameter	800 mm (31.5 inch)
Minimum wall thickness	8 mm (0.315 inch)
Test pressure, Ph	43 bar (624 psi)
Guaranteed min yield strength, Y	337 MPa (48,878psi)
Guaranteed min tensile strength, T	490 MPa (71,068 psi)
Wall stress at test pressure	Must be less than 0.75xY and 0.5xT
Minimum burst pressure	2.25 x Ph = 96.8 bar (1404 psi)

¹The requirements for this steel are specified in ISO 9328-3. The closest US-equivalent to this steel, for reference only, appears to be ASTM A737 Grade B.

b. REQUALIFICATION: Each pressure drum must be periodically inspected and requalified at least once every 5 years in accordance the visual inspections (external and internal) and pressure test (at 43 bar) procedures outlined in ISO 23088:2020.

c. OPERATIONAL CONTROLS:

(1) Pressure drums must be shipped in accordance with the provisions of § 173.40(c) for valve closures and § 173.40(d) for handling protection.

(2) Each pressure drum shall be tested for leakage after filling.

(3) The filling ratio must be in accordance with Packing Instruction P200 of the United Nations Recommendations on the Transport of Dangerous Goods, Model Regulations such that the outage and pressure requirements of § 173.40(b) are not violated.

(4) The shipping paper must identify the pressure drums with the special permit number “DOT-SP 21290”.

(5) Orion Engineered Carbons GmbH or Orion Engineered Carbons LLC must be the shipper of empty pressure drums. Empty and filled pressure drums must be transported under exclusive use by contract motor carriers that meet Federal Motor Carrier Safety Administration requirements. Gaylord Chemical Company, LLC must package the product in Orion Engineered Carbons LLC owned pressure drums and prepare for transport. The shipping preparation consists of the pressure drums being marked and labeled in accordance with the HMR and the pressure drums being placed/strapped to a cradle, which is then strapped to the shipping container.

(6) The pressure drums must be transported in a closed transport vehicle or freight container, which must be marked in accordance with § 172.313.

(7) Pressure drums may not be fitted with pressure relief devices. Emergency response information shall be provided with the shipment and available via an emergency response telephone number to indicate that the drums are not fitted with pressure relief devices and provide appropriate guidance for exposure to fire.

(8) Not more than 72 shipments of pressure drums are authorized for shipment per year from the USA to Germany with each shipment containing not more than 11 pressure drums.

(9) Pressure drums authorized under this special permit may be transported by cargo vessel only when loaded into closed cargo transport units.

(10) Each pressure drum authorized under this special permit may only be filled in the United States for export only and, be returned to the United States (with residue of hazardous material), for the purposes of refilling for export only.

8. SPECIAL PROVISIONS:

a. A person who is not a holder of this special permit who receives a package covered by this special permit may reoffer it for transportation provided no modification or change is made to the package and it is reoffered for transportation in conformance with this special permit and the HMR.

b. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

c. MARKING: Each drum shall be permanently and legibly marked "DOT-SP 21290" on a contrasting background. The marking shall be in accordance with the requirements of ISO 13769 as shown in CHEMET S.A. marking document No.4-1855-04/K on file with the OHMS.

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle and cargo vessel.

10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel or motor vehicle used to transport packages covered by this special permit.

11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) —“The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: RS/NICKS